Ganitha Piyasa: Effective Lesson Delivery Method for Graphical Dyscalculia Students

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***Abstract*—Many children face learning disabilities, each requiring unique solutions such as physical exercises, teaching techniques, and behavioral interventions. A specific challenge lies in Graphical Dyscalculia, which hinders children's mathematical abilities. This study presents the "Ganitha Piyasa" mobile application, tailored to help children with Graphical Dyscalculia enhance their numerical proficiency and number writing skills. The application provides a structured learning environment with fifteen progressively challenging levels. It offers performance-based feedback and evaluation, alongside comprehensive data collection recording task-specific details, completion times, and accuracy. An Artificial Neural Network (ANN) model, initially trained with the MNIST handwritten digits data set in Keras, accurately assesses number writing. Converted into a Tensor Flow Lite file, this model ensures efficient real-time evaluation. A Random Forest Classifier, utilizing metrics like average writing time and success probabilities, predicts a child's improvement in number writing skills. This aids personalized learning pathways. The study involved 50 children who were diagnosed with Dyscalculia aged seven to ten, demonstrating the application's effectiveness. "Ganitha Piyasa" boasts an intuitive interface that encourages engagement and parental involvement. Parents can monitor their child's progress through earned achievement stars, motivating sustained usage. The application's methodology aligns with medical expertise, ensuring effective enhancement of children's numerical skills.**

***Index Terms*—Dyscalculia, Graphical Dyscalculia, Mobile application, lesson delivery app**

I. Introduction

The term "learning disabilities" (LDs) is not very friendly for developing nations like Sri Lanka. Many pupils struggle with their learning because they are unaware of and uninformed about learning difficulties.[1] Students with learning difficulties are typically denied access to higher education and ignored when it comes to meeting learning objectives.[2] To provide a comprehensive understanding of the prevalence of Dyscalculia and its impact on the student population, we conducted a statistical analysis. We surveyed a sample of 500 students aged 7-10 from various educational institutions. Among them, we identified the presence of Dyscalculia and categorized it into three main types: Speech Dyscalculia, Memory Dyscalculia, and Graphical Dyscalculia[3]. The results of our analysis revealed that Dyscalculia affected a substantial proportion of the student population. Of the surveyed students,11% exhibited symptoms of Dyscalculia, with Speech Dyscalculia accounting for 4%, Memory Dyscalculia accounting for 5%, and Graphical Dyscalculia accounting for 2%. Further analysis of the data unveiled interesting trends within each Dyscalculia type. Among students with Speech Dyscalculia, 60% also exhibited developmental language disorders, highlighting the co-occurrence of these conditions. Memory Dyscalculia presented challenges in different areas, with 45% of affected students struggling with counting numbers and 35% facing difficulties in telling time. In Graphical Dyscalculia, the majority of students (70%) encountered problems with writing mathematical symbols and numerals[4]. Dyscalculia is a mathematical impairment that affects between 3% and 6% of children, according to estimates [5]. Counting things, classifying mathematical signs (addition, subtraction, multiplication, and division), playing games with numbers, and memorizing multiplication tables are all tough tasks that are associated with it [6]. Utilizing this statistical analysis, we aim to develop an interactive lesson delivery app that directly addresses the distinct needs of students with Dyscalculia. By tailoring our app's features to the specific Dyscalculia types,

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we aim to provide effective solutions and facilitate improved mathematical performance in this student population.

II. Literature Review

Mathematics is a fundamental subject that plays a crucial role in various aspects of life and education. However, for some students, learning mathematics can be a challenging endeavor due to specific learning difficulties. Dyscalculia, a type of learning difficulty characterized by poor math performance, affects a significant portion of the student population. Extensive research has been conducted to understand the prevalence, characteristics, and types of Dyscalculia, as well as to develop effective interventions to support students with this learning disorder. In considering existing applications in Sri Lanka,[9] The ”Pubudu” app is a deep learning and machine learning-based mobile application developed to screen and intervene in dyslexia, dysgraphia, and dyscalculia, affecting around 10% of children worldwide. It supports local languages in Sri Lanka and follows clinical screening procedures recommended by health professionals. The app achieved high screening accuracy, such as 58% for dyslexia, 88% for letter dysgraphia, 99% for numeric dysgraphia, and 90% for dyscalculia. Using gamified environments, ”Pubudu” motivates and empowers underprivileged children in Sri Lanka to overcome their learning disabilities. A Deep Learning Model to Predict Student Learning Outcomes was Developed in [10] The researchers conducted a study to explore how certain Key Performance Indicators (KPIs) based on student interactions with the Learning Management System (LMS) Blackboard could be used to predict student learning outcomes in higher education institutions. Blackboard is widely utilized in universities worldwide due to its capabilities in managing educational activities, content, interactions, and assessments.

The study utilized a mixed-methods approach, combining documentary analysis and deep learning models to predict student performance. Data from seven general preparation courses were collected from Blackboard reports. The researchers analyzed the data to identify potential predictive KPIs associated with the electronic Blackboard report. [11] in this research, the focus is on dyscalculia, a learning disability that affects around 6-7% of individuals and is often associated with a growing failure rate among pupils in mathematics. Dyscalculia is not well-known, which makes it challenging to identify and address, leading to negative impacts on everyday life. Early intervention is crucial to mitigating the severity of dyscalculia, but currently, there are no standard tests to detect it, and its origins remain poorly understood. Additionally, there is a significant gap in Portuguese special education, as there are no learning support systems for dyscalculia in the Portuguese language. [12] This is to conduct a reinforcement learning framework for teaching dyslexic students researchers have developed an adaptive reinforcement learning framework called RALF using Cellular Learning Automate (CLA) to generate content for students with dyslexia automatically. Dyslexia is a learning disorder that significantly affects reading abilities. Previous studies have emphasized the importance of using machine learning techniques in content supplements to adapt course concepts to learners' educational levels, but there has been limited research applying machine learning methods for adaptive content generation.

RALF starts by generating simplified font models of the alphabet as online representations. The CLA structure learns each rule of character generation through asynchronous reinforcement learning cycles. The framework then allegorically generates Persian words, considering the state of each character to decide the cohesiveness of the alphabet and the cells' response to the environment. RALF can also generate long texts and sentences using an embedded word-formation algorithm, determining the spaces between words through the CLA neighboring states. Additionally, RALF offers word pronunciation, exams, and games to enhance the learning performance of individuals with dyslexia. A content analysis of research on the technology used for teaching mathematics to students with disabilities [13] researchers have developed an adaptive reinforcement learning framework called RALF using Cellular Learning Automate (CLA) to generate content for students with dyslexia automatically. Dyslexia is a learning disorder that significantly affects reading abilities. Previous studies have emphasized the importance of using machine learning techniques in content supplements to adapt course concepts to learners' educational levels, but there has been limited research applying machine learning methods for adaptive content generation.

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This research paper addresses the issue of learning disabilities, particularly focusing on dyslexia, dysgraphia, and dyscalculia among school children. The paper introduces a mobile application named 'Helply' that aims to assist children in improving their color identification skills, reading skills, and short-term memory skills through interactive and collaborative activities, while also reducing the impact of their dyslexic disorders. Additionally, the application utilizes an NAO Robot, which can take voice inputs using voice recognition technology and process images using image processing technology.

The paper highlights the fact that learning disabilities are often overlooked in treatment, possibly due to a lack of awareness of the risks and limited access to proper medical care. Since learning disabilities are considered neurological conditions with an unknown cause, there is no definitive therapy or cure available[14].

The research paper addresses the significant issue of

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dyslexia, a hidden neurological learning disability that affects students' ability to accurately and fluently recognize words and connect letters to their corresponding sounds. In India, there's a notable increase in the prevalence of dyslexia. The challenges faced by dyslexic individuals can vary, as their brains are typically normal and often quite intelligent, but their strengths might lie in areas other than language skills. This discrepancy can lead to low self-esteem, mood swings, negative emotions, and even depression among these students.

The paper emphasizes the critical importance of early detection and evaluation of dyslexic students. To this end, the authors have conducted a review that summarizes various research dimensions related to dyslexia detection. The central focus of this review is on the utilization of machine learning techniques for identifying dyslexia. These techniques encompass a range of approaches, including game-based methods, image processing techniques, and different machine learning-based assessments and assistive tools. These tools are designed to support and alleviate the challenges faced by dyslexic individuals[15].

There are many researchers who have developed mobile-based systems to help students in their studies and improve the student's knowledge [16][17].

study was to analyze the content of research focused on the use of technology for teaching mathematics to students with disabilities. The researchers conducted an analysis using word networks and structural topic modeling on a dataset of 488 studies published between 1980 and 2021.

The findings of the study revealed certain key trends over the years. In the 1980s and 1990s, the words "computer" and "computer-assisted instruction" held significant importance, indicating a strong emphasis on technology in teaching. During the 2000s and 2010s, the term "learning disability" emerged as a central concept, reflecting the growing attention to addressing the needs of students with disabilities[18].

To overcome the problems and limitations in the currently done research and developed apps "Ganitha Piyasa" is built to explore the effectiveness of these approaches and develop a comprehensive approach to teaching mathematics to age seven to ten children with dyscalculia.

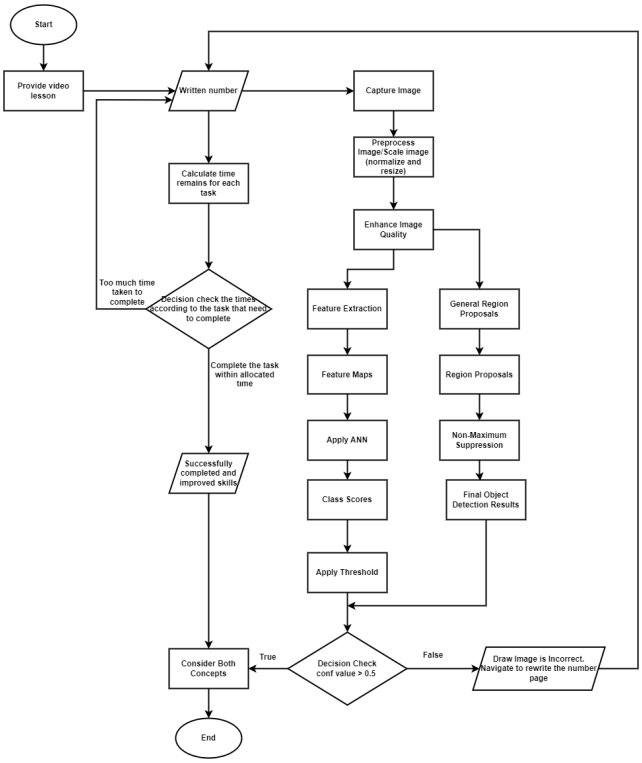


Fig. 1: Flow Chart of the application



Fig. 2: System Diagram

III. METHODOLOGY

The mobile application or lesson delivery system for children with Graphical Dyscalculia should meet several functional requirements. Firstly, it should be able to collect and store data on the performance of these children. This data should include information about specific tasks or activities, the time spent on each task, and the child's level of accuracy.Fig.1 and Fig.2 show the flow chart and system diagram for the system respectively.

First, every child needs to log in to the "Ganitha Piyasa" mobile application. for that children need their parent's support. with their support children can access the application. Fig.3 shows the interfaces. Next, they can select a writing numbers game in the task list to improve their number writing ability.Fig.3(f) shows the interfaces. First, select level one, it provides a lesson on how to write the number zero after watching that video lesson, the user is asked to write the number zero. likewise, level one to nine numbers, and then the user was asked to write numbers from zero to nine with the guide. children have more freedom to use the given space. in the next five levels provide the image with the number of objects in the image and it should write the canvas. All fifteen levels have three stars to achieve. If a percentage of 25% is obtained, it achieves one star, If a percentage of 50% is obtained, it achieves two stars, and If a percentage greater than or equal to 70% is obtained, it achieves all three stars. only when each level archives at least two stars will the next level unlock. at the end of each level, the children's level is displayed then parents can get an idea about how their child

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improved.

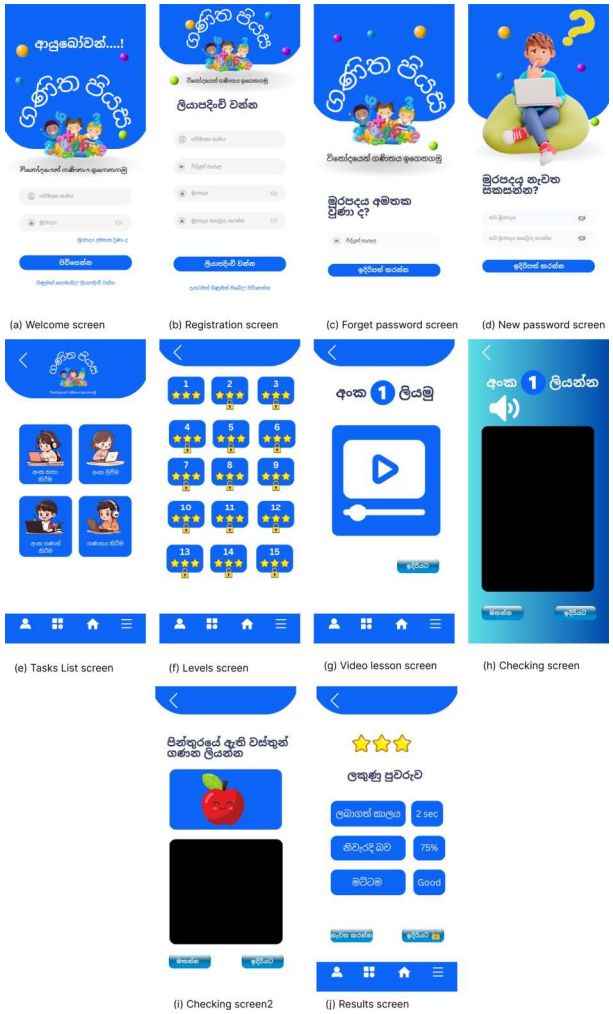


Fig. 3: Sample Interfaces

To evaluate graphical dyscalculia children's following parameters are taken[7]; accuracy, and completion time. A trained ML model is used to evaluate the accuracy of the number. This model makes use of the ANN-containing MNIST handwritten digits dataset[8]. The ANN consists of 19 layers, and 20 epochs were employed to achieve its high accuracy of 98.3%. After that, the numerical model was transformed into a Tensor Flow Lite file. After the number writing task is finished, the handwriting is evaluated. A Random Forest Classifier is constructed using the parameters of average writing time, and success probabilities of written numbers. The model will forecast the likelihood that a child's ability to write and improve. To train the model, a total of 50 children, who have been diagnosed dyscalculia are used. The subjects include children between the ages of seven to ten who attend the "Ayathi" National Center Ragama. Fig.3(j) shows whether the child is at the level of a normal child of that age, or if the child is at a lower level. This application aims to improve the dyscalculia child to the level of a normal child using the "Ganitha Piyasa" mobile application. The medical experts who devised the intervention game made sure that it helped kids improve their skills.

IV. EXPERIMENT AND RESULT

Results obtained when training the model to identify written input are included in this section. Results obtained in the process are shown in Table 1.

Table 1: Results of trained model

Classification Accuracy	Loss	Validation Accuracy
98.28%	6.38%	96.75%

The model with the most classification and validation accuracy was chosen for integration with the system to identify handwritten numbers. The results of the chosen model's training data are shown in Fig.4 and Fig.5.

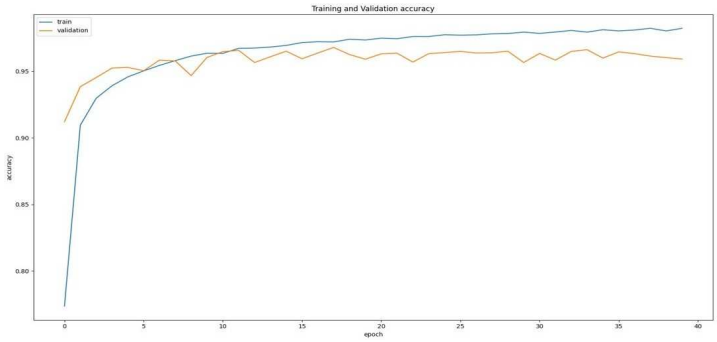


Fig. 4: Training & validation accuracy of model

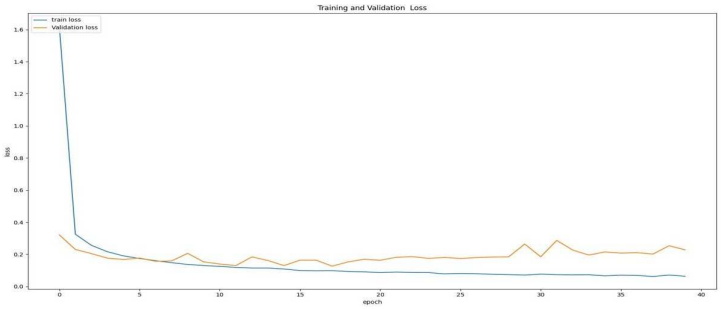


Fig. 5: Training accuracy & loss

When the tasks completed after teaching all the lessons satisfy the children's progress of the lessons. After learning the lessons carefully, students attempted simple tasks of them regarding the lesson. When they learn the write zero to nine numbers, they face simple similar number writing tasks. for the number, children face two types of tasks 1: writing the correct number for the guide video. 2: Identify the correct object count in the displayed image and write the correct count. In this approach, we increase the manual effort for the children to write the numbers, because after watching the lesson they may forget in the next season, supposed to give the chance for them to write the answer and upload the answer. In answer

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evaluation, the system processes the correct answer through the Tensor Flow Lite file.

V. CONCLUSION

”Ganitha Piyasa” is Sri Lanka’s first Sinhala medium mobile application that teaches mathematics exclusively to children with Dyscalculia. The success of this research is shown in Dyscalculia because of the extremely precise teaching methods used. We intend to add more data to our model to improve accuracy. Due to the recommendation of health professionals that children should receive at least eight months of instruction through the use of applications, the intervention of these children is still in progress. Though ”Ganitha Piyasa” supports only the Android platform now, in the future it will be upgraded to support other operating systems. Further ”Ganitha Piyasa” will be extended to support other different languages such as English, and Tamil and it will support other languages in other countries with the support of the health ministry. We plan to use ”Ganitha Piyasa” to spread this across the country to produce children who are talented in mathematics. Our goal is to improve our mobile application so that no child is weak in math.

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